Current base force model

$$\vec{b}(p) = S_i \cdot \left({}^B_I\hat{g}_I\, K_I \cdot I\right) \qquad \text{其中} \quad \begin{cases} {}^B_I\hat{g}_I = \dfrac{6}{5}k_{11} \cdot \dfrac{N_c}{4\pi\hat{\ell}^2 R_a} \\ S_i = \dfrac{\bar{p}_i - \bar{p}_c}{\|\bar{p}_i - \bar{p}_c\|^3} \end{cases}$$

$$\vec{m}(p) = \left(\frac{3V}{\mu_0}\right)\left(\frac{\mu - \mu_0}{\mu + 2\mu_0}\right) \cdot \vec{b}(p) = {}^M\hat{g}_B \cdot \vec{b}(p)$$

$$\vec{F}(p) = \frac{1}{2}\nabla(\vec{m} \cdot \vec{b}) = \frac{1}{2}\nabla\left({}^M\hat{g}_B\, \vec{b} \cdot \vec{b}\right) = \frac{1}{2}{}^M\hat{g}_B\, \nabla\left(\|\vec{b}\|^2\right)$$

$$= \underbrace{\frac{{}^M\hat{g}_B\, {}^B\hat{g}_I^2}{2\hat{\ell}}}_{{}^F\hat{g}_I} \left(I^T K_I^T \begin{bmatrix} L_x(\bar{p}) \\ L_y(\bar{p}) \\ L_z(\bar{p}) \end{bmatrix} \cdot K_I \cdot I \right)$$

$$= {}^F\hat{g}_I \left(I^T K_I^T \begin{bmatrix} L_x(\bar{p}) \\ L_y(\bar{p}) \\ L_z(\bar{p}) \end{bmatrix} K_I \cdot I \right) \qquad \text{其中} \quad \begin{cases} L_x(\bar{p}) = \dfrac{\partial}{\partial x}\left(S_i^T S_i\right) \\ L_y(\bar{p}) = \dfrac{\partial}{\partial y}\left(S_i^T S_i\right) \\ L_z(\bar{p}) = \dfrac{\partial}{\partial z}\left(S_i^T S_i\right) \end{cases}$$

Voltage base force model

$$\vec{F} = \underbrace{\frac{{}^M\hat{g}_B\, {}^B\hat{g}_V^2}{2\hat{\ell}}}_{{}^F\hat{g}_V} \cdot \left(V^T \bar{D}^T \begin{bmatrix} L_x(\bar{p}) \\ L_y(\bar{p}) \\ L_z(\bar{p}) \end{bmatrix} \bar{D}\, V \right) = {}^F\hat{g}_V \left(V^T \bar{D}^T \begin{bmatrix} L_x(\bar{p}) \\ L_y(\bar{p}) \\ L_z(\bar{p}) \end{bmatrix} \bar{D}\, V \right)$$